ACE Cyber Research Question & Hypothesis: Fall 2025

1. 1. Project Information

Student Name(s): Denzel Jefferson_____

Project Title: Contextualized Threat Intelligence Prioritization for CTI Analysts_____

Date / Version: May 4, 2026 _____

Mentor/Advisor: ___Professor William Littleton_____

2. Research Question

State the central question your research project aims to answer (be specific and measurable):

How can operational environment data gathered from Elasticsearch be used to automate and improve threat intelligence prioritization within CTI workflows?_____
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3. Hypothesis

Formulate a testable hypothesis based on your research question and your completed literature review. This should follow the material covered in session #1.

If operational environment data from Elasticsearch is incorporated into an automated MISP-based threat-prioritization system, then the time required for CTI analysts to identify high-priority threats will significantly decrease while maintaining or improving overall prioritization accuracy compared to a manual review process.

5. Variables / Factors to Consider

Identify key variables, factors, or conditions relevant to your hypothesis. For qualitative research, list constructs or themes.

- Independent Variable(s): Contextualized threat intelligence prioritization system that will incorporate Elasticsearch environment data.
- Dependent Variable(s): Time needed to identify high-priority threats, Reduction in CTI analyst alert fatigue, and Accuracy of threat prioritization (High, Medium, Low).
- Control/Contextual Factors (if any): The same dataset is used to test automated and manual review. The same testing environment and Elasticsearch-derived operational

data are used across all test scenarios. The same simulated workload will be used to assess the effectiveness of automated prioritization compared with manual review._____

6. Rationale

Explain why you expect this hypothesis to hold true. What theory, evidence, or reasoning supports it?

CTI analysts are currently overwhelmed by the large volume of raw, unfiltered threat intelligence data, which slows response times and causes alert fatigue. When alerts appear, they may be low priority, but they still require thorough manual review. By using operational environment data from Elasticsearch, we can implement an automated prioritization system that ranks threats by their relevance to the environment. This will allow CTI analysts to focus solely on high-priority threats instead of sorting through numerous alerts. Since the system applies weighted contextual scoring to rank threat relevance, it is expected to significantly reduce the time required to identify high-priority threats while improving overall analyst efficiency compared to manual review.